

Weekly report: Theoretical framework for EIT in NV centers

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Question

How much is the theoretical upper temperature limit for realizing EIT in NV centers?

Hypothesis

It is possible to expand the temperature for realizing EIT in NV centers to Room temperature by optimizing on-axis magnetic field, B_z and off-axis crystal strain, $\delta^\perp$ to increase the energy difference between the target three-level system and nearby excited states.

Evidence

(1) Phonon-induced decoherence in NV centers: Γ_{XY} increases rapidly with temperature, following a T^5 dependence, which significantly raises the required control-field Rabi frequency (Ω_{contrast}) for achieving EIT at higher temperature, and this increase in Ω_{contrast} can surpass the energy difference (Δ_e) between the target three-level system and nearby excited states, leading to unwanted coupling and breakdown of EIT (Figure 1).

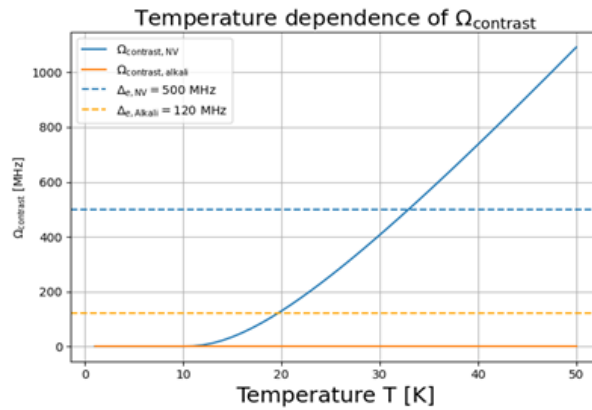


Figure 1: Comparison of Ω_c dependence of EIT contrast in NV centers and alkali atoms at different temperatures.

(2) However, by optimizing the conditions of the magnetic field B_z and the perpendicular crystal strain $\delta^\perp$, it is possible to sufficiently increase Δ_e (Figure 2).

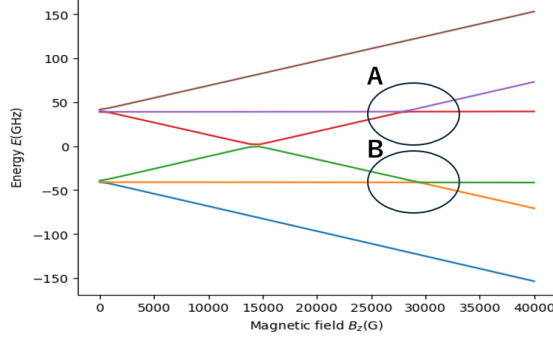


Fig 2: Difference in energy levels between different orbits in NV centers ($\delta^\perp = 70\text{GHz}$).

Gap

(1) The condition for realizing EIT in NV centers can be quantitatively expressed as equation(1), where γ_{21} is the decoherence rate between the two lower levels of the three-level system, approximately equal to the inverse of T_2 ($\cong 1\text{MHz}$). At room temperature ($T = 300\text{K}$), $\sqrt{\Gamma_{XY}\gamma_{21}}$ is approximately 7.8GHz .

$$\sqrt{\Gamma_{XY}\gamma_{21}} < \Omega_{\text{contrast}} < \Delta_e(B_z, \delta_\perp) \quad (1)$$

(2) At $\delta^\perp = 70\text{GHz}$, the Λ systems formed by $|E_x, m_s = 0\rangle$ and $|E_y, m_s = +1\rangle$ (Circle A), and by $|E_x, m_s = -1\rangle$ and $|E_y, m_s = 0\rangle$ (Circle B) have an energy difference Δ_e of approximately 50GHz , which is significantly larger than the required Ω_{contrast} for EIT at room temperature. This suggests that with proper optimization of B_z and $\delta^\perp$, it may be possible to achieve EIT in NV centers at temperatures higher than currently demonstrated limits.

Next step

(1) To identify the optimal conditions for achieving EIT in NV centers, a systematic numerical analysis should be conducted to explore the parameter space of B_z and $\delta^\perp$, and to evaluate the resulting Δ_e and its impact on the EIT contrast at various temperatures.